

matrix invertible? Justify your answer.

Solution: A square lower triangular matrix is invertible if and only if each of its main diagonal entries are non-zero. The reasoning is similar to Problem 13.

15) Can a square matrix with two identical columns be invertible? Why or why not?

Solution: Let A be a $n \times n$ square matrix, where $A = [a_1 \ a_2 \ \dots \ a_n]$, $a_i \in \mathbb{R}^n$, $1 \leq i \leq n$. Let a_i and a_j be two arbitrary columns such that $a_i = a_j$. WLOG, $i < j$.

$$\therefore a_j = 0 \cdot a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_{i-1} + 1 \cdot a_i + 0 \cdot a_{i+1} + \dots + 0 \cdot a_{j-1} + 0 \cdot a_{j+1} + \dots + 0 \cdot a_n.$$

By defn. of linear combination, a_j is a linear combination of the rest of the columns of A . By Theorem 7 from Chapter 1, the set of columns of A are linearly dependent.

By the Invertible Matrix Theorem, the columns of A form a linearly independent set if and only if A is an invertible matrix.

$\therefore A$ is non-invertible. $\square$

16) Is it possible for a 5×5 matrix to be invertible when its columns do not span $\mathbb{R}^5$? Why or why not?

Solution: Given, a square 5×5 matrix B whose columns do not span $\mathbb{R}^5$. By the Invertible Matrix Theorem and the equivalence of the statements (a) and (h) we know that a square $n \times n$ matrix A is invertible if and only if the columns of A span $\mathbb{R}^n$. $\therefore$ The matrix B is not invertible. $\square$

17) If A is invertible, then the columns of A^{-1} are linearly independent. Explain why.

Solution: Given, A is invertible. By defn, there exists a unique matrix A^{-1} s.t. $AA^{-1} = A^{-1}A = I_n$, assuming A is a square $n \times n$ matrix. By Theorem 6(a), A^{-1} is invertible. By the Invertible Matrix Theorem and the equivalence of the statements (a) and (e),